# Analyze wellness attributes

## Active IQ Digital Advisor

NetApp
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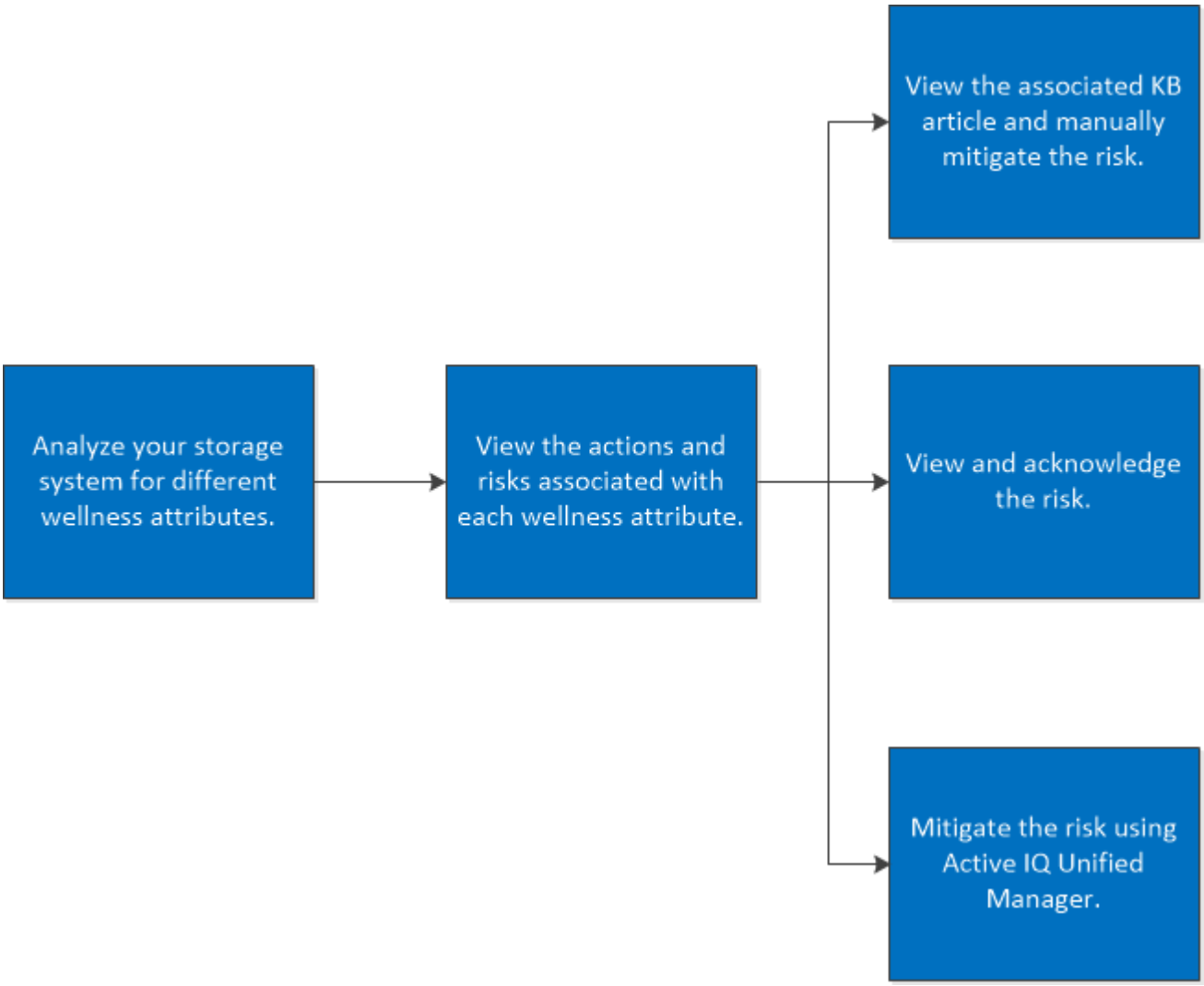
# Analyze wellness attributes

## Understand wellness

Wellness widget provides detailed information about your storage system. It provides information about different attributes of your storage system, such as performance and efficiency, capacity, configuration settings, security, renewals, and others.




The wellness widget also provides information about the risks and the actions that should be taken to mitigate the risk for each wellness attribute. The following are the types of risks and the associated consequence for each risk:

| Risk Type | Consequence |
|---|---|
| Critical | Data loss, data corruption, cluster data outage, personal safety issue, or potential legal compliance issue. |
| High | Short-term loss of data access or prolonged loss of node redundancy. |
| Medium | Performance degradation or short-term loss of node redundancy. |

| Risk Type | Consequence |
| --- | --- |
| Low | Low impact scenarios |
| Best Practice | Deviations from documented best practices |

You can view the following video to understand the importance of the wellness attributes:

## View the risk and manually take corrective actions

You can analyze the wellness attributes of your storage system by viewing the actions and risks associated with them. You should view the associated corrective actions and manually mitigate the risk.

**Steps**

1. Click the **Wellness** widget on the dashboard or click **View All Actions** to view the list of all the actions and risks.
2. View the **Actions** and **Risks** associated with the wellness attribute.
3. Click **Actions** to view the risks associated with the actions, click **Risks** to view all the risks, or click **Affected Systems** to view the systems that require attention.
4. Click the risk name to view information about the risk.
5. Click the associated corrective actions and follow the information to resolve the risk.

The steps to mitigate the risks are same for all wellness attributes. You can view the following video to monitor and fix security related issues:

# Increase protection against hackers and Ransomware attacks

When you log in to the Active IQ Digital Advisor and notice the red badge on the **Security** widget, you can take actions to fix critical risks.

**Steps**

1. Log in to Active IQ Digital Advisor.
2. Click **Actions** in the **Security** widget.

For the high-impact security risks, follow the recommended action that is to upgrade the operating system.

3. Click the **Unique Risks** tab, and then click the link in the **Corrective Action** column.

| Fix It | Risk Name | Mitigation | Corrective Action | Systems | Impact |
|---|---|---|---|---|---|
| | Clustered Data ONTAP has been determined to ha... | Potentially Non-disruptive | NTAP-20180423-0003 | 1 | High |

The risk is fixed in ONTAP 9.7P8 and later.

**Advisory ID:** NTAP-20200814-0005 **Version:** 6.0 **Last updated:** 12/03/2020 **Status:** Interim. **CVEs:** CVE-2020-9490, CVE-2020-11984, CVE-2020-11993

Overview Affected Products Remediation Revision History

Software Versions and Fixes

NetApp's currently available patches are listed below.

| Product | First Fixed in Release |
| --- | --- |
| Clustered Data ONTAP | https://mysupport.netapp.com/site/products/all/details/ontap9/downloads-tab/download/62286/9.5P15<br>https://mysupport.netapp.com/site/products/all/details/ontap9/downloads-tab/download/62286/9.6P11<br>https://mysupport.netapp.com/site/products/all/details/ontap9/downloads-tab/download/62286/9.7P8 |

4. The most important step is to plan OS upgrade in the **Upgrade Advisor** in Active IQ Digital Advisor.

# View and acknowledge the risk

You can analyze the wellness attributes of your storage system by viewing the actions and risks associated with them. You should view the corrective actions and manually mitigate the risk.

**Steps**

1. Click the wellness attribute widget on the dashboard or click **View All Actions** to view the list of all the actions and risks.
2. View the **Actions** and **Risks** associated with the wellness attribute.
3. Click **Actions** to view the risks associated with the actions, click **Risks** to view all the risks, or click **Affected Systems** to view the systems that require attention.
4. Click the risk to view the risk summary.
5. Click **Ack** to acknowledge the risk.

   The detailed risk summary information is provided along with corrective actions that should be manually performed to mitigate the risk.

6. If you do not want to or are unable to mitigate the risk at this time, provide the values for the fields and click **Acknowledge**.

   The risk will be added to acknowledged risks.

> If you no longer want a risk to be acknowledged, you can disregard the risk by clicking **Un-Ack** and following the same steps.

# View wellness history

You can view system risks occurring in the past three months, so that you can learn how they are faring overtime.

These risks are classified under four types of risks— **Unresolved**, **New**, **Resolved**, and **Acknowledged**. They are represented by different colors. The summary of these risks is represented through a **Risk History** graph.

**Steps**

1. On the dashboard, in the **Wellness** pane, click **View All Actions**.
2. Click **Wellness History**.
3. In the **Risk History** graph, click the category for which you want to view the risk history.

   When you hover over the colored bars, they display information on the number of risks in each category. Upon clicking the respective risk category, the information gets displayed in the **Risk Information** table.

   You can also download risk summary in an Excel sheet.

# View risks that can be automatically mitigated using Unified Manager or Ansible Playbook

You can analyze your storage system by viewing the actions and risks, and mitigate them using Active IQ Unified Manager or Ansible Playbook.

**Steps**

1. Click **View All Actions** on the dashboard.
2. Click **Actions** to view the risks associated with the actions, click **Risks** to view all the risks, or click **Affected Systems** to view the systems that require attention.

   If the risk can be mitigated using Active IQ Unified Manager, the icon is highlighted and if the risk can be mitigated using Ansible Playbook, the icon is highlighted.

| To mitigate the risk using Unified Manager | To mitigate the risk using Ansible Playbook |
| --- | --- |
| 1. Click the icon.<br>2. Click **Fix It** to launch Active IQ Unified Manager.<br>3. Click **Install** to install Active IQ Unified Manager 9.7 or later to use the **Fix It** option.<br>4. Click **Upgrade** to upgrade to Active IQ Unified Manager 9.7 or later to use the **Fix It** option. | 1. Click the icon.<br>2. Click **Download** to download the AFF and FAS firmware Ansible Automation package. |

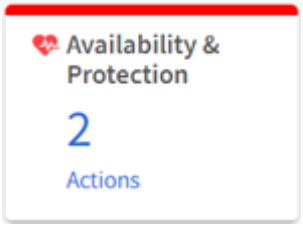
A SupportEdge Advisor or SupportEdge Expert contract is required to use the **Fix It** option and the Ansible Playbook features.

# Avoid the downtime and possible data loss

When you log in to the Active IQ Digital Advisor and notice the red badge on the **Availability and Protection** widget, you can take actions to fix critical risks. Without the firmware fix, these drives are vulnerable to become inoperable after a certain number of hours of being powered on. Fixing this would avoid both the downtime and possible data loss.

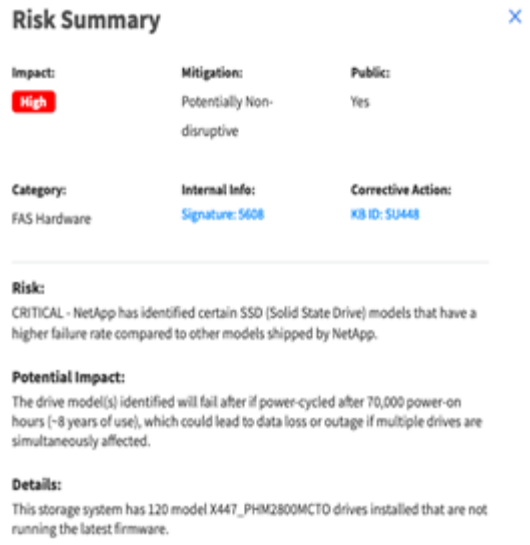
**Steps**

1. Log in to the Active IQ Digital Advisor.
2. Click **Actions** in the **Availability & Protection** widget.




   For the high-impact security risks, follow the recommended action that is to update disk firmware.

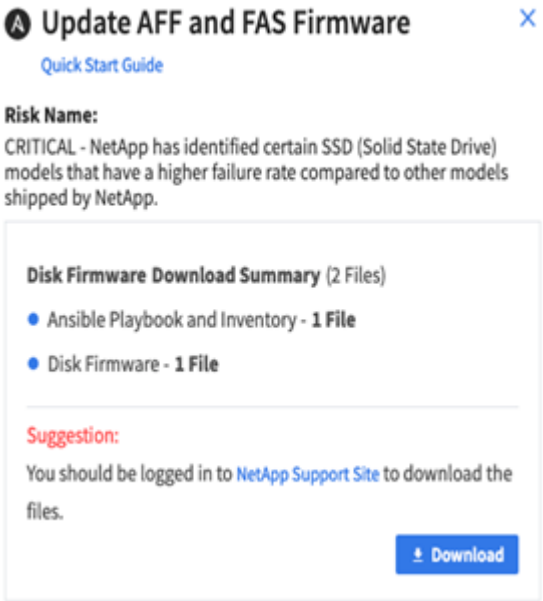
3. Click the **Risk Name** link for viewing risk summary.




4. Click the **Unique Risks** tab, and then click the link in the **Corrective Action** column.

Active IQ Digital Advisor generates custom Ansible scripts or playbooks to update the required disk firmware, including the disk firmware files.

5. Click the **Ansible "A"** icon to generate and download the scripts.




## Subscribe to wellness review email

You can subscribe to the wellness review email to receive a monthly email that summarizes wellness status, systems that are nearing their renewal dates, systems that require an upgrade for the NetApp products in your installed base.
You will receive a wellness review email so that you can view a monthly summary and

take action for your systems.
You also have options to view, edit, share, and delete your subscriptions. At any time, if you decide to not receive the email, you can unsubscribe from getting email as well.

After the subscription is enabled, you should select a start date when adding a subscription.
The monthly email summary provides a view of outstanding wellness, renewal, upgrade, and health check actions. You can confirm the email address and the email is sent to the specified email address.
You also have the option to delete subscriptions.

> This feature is available only through NetApp SupportEdge Advisor and SupportEdge Expert service offerings.

**Steps**

1. On the Health Check Assessment Dashboard, click **Wellness Review Subscription**.
2. Click **Add Subscription**.
3. Provide the required information in the **Name the Subscription**, **Choose Category**, **Search Customer**, and **Email** fields in the **Add New Subscription** dialog box.
4. Click **Subscribe**.

   Upon successful subscription, you will receive a **Subscription was added** message.

# Renew the software and hardware of your storage system

You can proactively identify the software and hardware that have expired or are near expiration in the next 6 months, and send a request to renew the hardware and software.

**Steps**

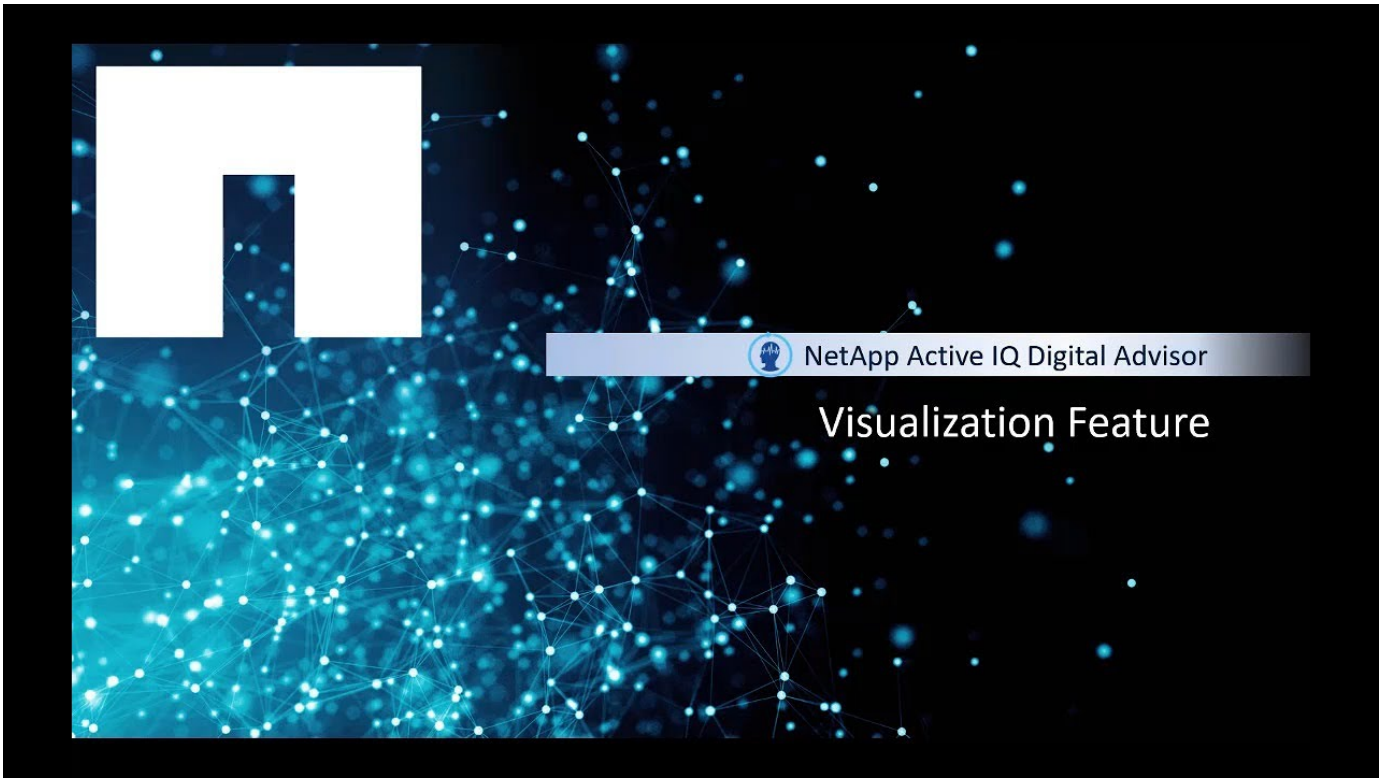
1. Click the **Renewals** widget on the dashboard or click **Renewals** from the **Planning** widget.
2. Select the systems that you want to renew and click **Renew**.
3. Optionally, provide additional comments.
4. Click **Send**.

# Analyze the wellness of clusters and nodes

You can analyze the wellness of your clusters and nodes using ClusterViewer, a one-stop source for information on the physical and logical configuration of your clusters and nodes.

ClusterViewer provides information, such as stack diagrams of your nodes, storage usage and efficiency, headroom in hardware capacity, and so on, that helps you take informed decisions to improve the wellness of your clusters and nodes.

You can view visualizations or graphical representations of the physical configuration of your nodes at cable, stack, and RAID Disk levels. You can also download the visualizations in SVG format.




**Steps**

1. In the **Inventory** widget, select the cluster or node (host) that you want.
2. At the cluster or node level, click **ClusterViewer** next to the **Configuration** widget.
3. Click the **Visualization** tab to view a graphical representation of the cluster.

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